Course Code	Course Title	L	T	P	C
MAENG501	Technical Report Writing	1	0	4	3
Pre-requisite	NIL	**Syllabus Version**			
		1			

Course Objectives

To develop communicative competence in students.
To apply advanced technical communication principles in practical writing tasks for diverse professional contexts.
To execute research and analytical skills to produce coherent and evidence-based technical documents through practical exercises.

Course Outcomes

Apply the principles of effective report writing.
Demonstrate the ability to draft and present technical reports.
Produce key components of technical documents, including sections focusing on purpose, audience, and structure, through writing assignments.
Create technical summaries and abstracts by practicing techniques for summarizing and paraphrasing complex information.

Module:1	**Fundamentals of Technical Writing**	**3 hours**
Introduction to Technical Writing: Definition and typical forms (reports, instructions, proposals); Key Factors in Technical Writing: Purpose, Audience, and Tone; General Writing Basics: Clarity, fluency, effectiveness; The Process of Writing: Pre-writing, Writing, and Post-writing stages; Organization in Writing: Having an outline, using introductions, headings, lists, figures, and summaries.		
Module:2	**Technical Grammar**	**2 hours**
Concord; Tense Shifts		
Module:3	**Introduction to Reports and Report Structure Front Matter**	**2 hours**
Reports in Organizations: Role and importance of reporting in corporate and industrial segments.; Purpose of Reports: Conveying decisions, facts, and information accurately and up-to-date.; Report Structure Overview: Division into Front Matter, Main Body, and Back Matter; Components of the Front Matter		
Module:4	**Report Structure Main Body and Back Matter**	**4 hours**

Components of the Main Body: Introduction, Discussion or Description, Conclusions, and Recommendations; Writing Introduction, Discussion or Description, Conclusions and Recommendations, Methods of Reporting: The Letter Method and The Schematic Method; Routine Reports: Nature, frequency, and function in organizations (often statistical, fixed intervals).		
Module:5	**Technical Proposals**	**2 hours**
Punctuation Right words and phrases; avoiding cliches, jargons, foreign words and phrases, ambiguity, redundancy, circumlocution - Developing hints		
Module:6	**Contemporary Issues**	**2 hours**
Guest Lecture		
	Total Lecture Hours:	**15 hours**
Text Book(s)		
Kumar. S & Pushplata. , "**Effective Communication Skills.** ", New Delhi OUP, **2018** Muralikrishna and Sunita Mishra, "**Communication Skills For Engineers**", Pearson, 2nd Edition, **2011** Shirley Mathew., "**Effective Communication Skills.**", Nirali Prakashan, **2025**		
Reference Books		
Indicative Experiments		
1. **Introduction to the Technical Writing** Introduction to Technical Report Writing**;** Analyzing and identifying the characteristics of effective technical writing in sample documents.; Short writing exercises focusing on clarity, conciseness, and identifying purpose, audience, and tone in simple technical scenarios. **Activity:** Reviewing and providing feedback on short technical descriptions.		2 hours
2. **Analyzing Audience and Context for Practical Writing** Applying reader and stakeholder analysis techniques to specific engineering report scenarios; Drafting content segments tailored for different technical and non-technical audiences (e.g., writing an executive summary for managers vs. a technical description for fellow engineers); Simulating audience needs assessment based on given project descriptions. **Activity:** Developing a 'reader profile' for a major report project.		4 hours
3. **Technical Grammar and Style in Practice**		2 hours

Intensive practical exercises on complex grammatical structures, sentence syntax, and common errors in technical writing; Exercises in applying principles of clarity, conciseness, and precision to improve technical sentences and paragraphs; Practicing the appropriate use of active and passive voice in different report sections. **Activity:** Peer-editing session focusing on grammar, mechanics of writing, and technical style.	
4. **The Writing Process and Report Outlining Lab** Practicing prewriting techniques for complex technical topics: brainstorming, mind mapping, and systematic outlining; Developing detailed hierarchical outlines for a major technical report project, including main and sub-points; Planning content organization based on report type (e.g., feasibility, empirical research, lab report) and audience needs. **Activity:** Group exercise to compare and refine report outlines.	4 hours
5. **Writing the Report Front Matter and Introduction** Detailed practical guidance on drafting all components of the report Front Matter; Creating a professional Title Page, Forwarding Letter/Preface, Acknowledgements, and Table of Contents for the ongoing report project; Drafting effective Abstracts and Summaries (Executive Summaries) based on provided technical content, focusing on capturing the essence. **Activity:** Writing the Introduction section of the report project, including background, scope, and objectives.	4 hours
6. **Writing the Report Body: Data, Descriptions, and Discussion** Techniques for presenting data, technical descriptions, and analysis in the main body; Drafting sections of the report body focusing on presenting organized data and technical details; Writing the "Discussion" section, focusing on interpreting results and explaining findings based on provided or self-generated data. **Activity:** Integrating data and analysis points into the draft of the report body.	4 hours
7. **Writing the Report Body: Conclusions and Recommendations** Practical methods for drawing logical conclusions and formulating actionable recommendations; Drafting the "Conclusions" section based on the data and	4 hours

discussion from Module 6; Developing clear and practical recommendations based on the conclusions, considering the report's purpose and audience. **Activity:** Writing the "Recommendations" section of the report project.	
8. **Transcribing Visuals** Using charts, graphs and tables; Transcribing visuals that are clear, accurate, and effectively support the report's text; Integrating created visuals into the report draft, ensuring proper placement, captions, and referencing within the text. **Activity:** Peer review focusing on the effectiveness and integration of visuals.	2 hours
9. **Report Back Matter, Condensation, and Final Review** Practical session on creating the Back Matter: Appendices, Bibliography, Glossary, and Index (if applicable). Emphasis on consistent citation and referencing styles; Compiling Appendices and formatting a Bibliography for the report project; Practicing techniques for précis writing and summarization to condense longer texts. **Activity:** Final review of the complete report draft, focusing on overall structure, flow, formatting, and coherence.	2 hours
10. **Presentation Skills** Short Presentations, Formal Presentation with PPT Analytical Presentation of Charts, Graphs and Tables **Activity**: Presentations – Individual and Group	2 hours
Total Laboratory Hours:	**60 hours**
Mode of Evaluation : Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Lab Continuous Assessment, Lab Final Assessment, Presentaion	
Recommended by Board of Studies :	16-05-2025
Approved by Academic Council : No. 78	12-06-2025